CS 5402: Introduction to Data Mining

SP2026

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Missouri S&T

https://hyang428.github.io/

MISSOURI S&T

Course Schedule and Contacts

- Course time:
  - 2:00pm - 3:15pm, Tuesday and Thursday
- Course room:
  - Computer Science 222
- Instructor:
  - Huiyuan Yang
- Email:
  - hyang@mst.edu
  - Email with subject starts with: CS5402-SP26: xxxxxxx (to make sure your email is picked by my filter rules).
- Office hours:
  - CS 333 (3:20-4:20pm, Tu, Th) or by appointment
- TA:
  - TBA
  - Office hours: TBA

Course Philosophy

How This Course Is Designed

You will learn:

- How algorithms work (intuition + math)
- When to use which method
- How to apply them to real problems

Balance:

- Theory
- Practice
- Applications

Tools & Environment

What Tools Will We Use?

- Python
- NumPy / Pandas
- scikit-learn
- Visualization tools (i.e., Matplotlib, Seaborn)
- Pytorch

No prior ML library expertise required.

What You Should Be Able to Do

Learning Outcomes

By the end of this course, you should be able to:

- Identify data mining problems
- Choose appropriate methods
- Implement solutions
- Interpret results critically

Why Data Mining Matters

Why Do We Need Data Mining?

Large-scale Data is Everywhere!

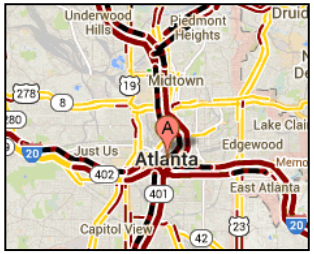
- There has been enormous data growth in both commercial and scientific databases due to advances in data generation and collection technologies
- New mantra
  - *Gather whatever data you can whenever and wherever possible.*
- Expectations
  - Gathered data will have value either for the purpose collected or for a purpose not envisioned.



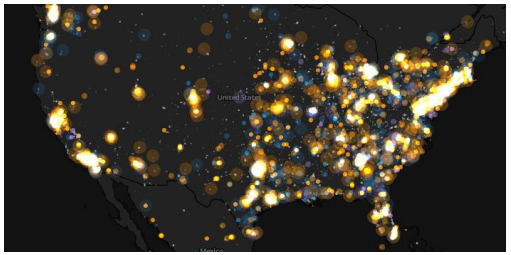
Cyber Security



E-Commerce



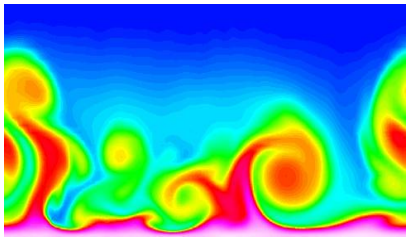
Traffic Patterns



Social Networking: Twitter



Sensor Networks



Computational Simulations

Why Data Mining? Commercial Viewpoint

- Lots of data is being collected and warehoused
  - Web data
    - Google has Peta Bytes of web data
    - Facebook has billions of active users
  - purchases at department/ grocery stores, e-commerce
    - Amazon handles billions of visits/month
  - Bank/Credit Card transactions





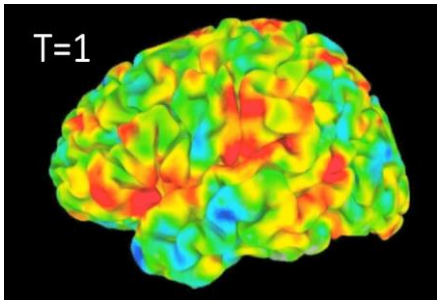




- Competitive Pressure is Strong
  - Provide better, customized services for users (e.g. in Customer Relationship Management, recommend products/videos, etc) can attract more users and generate higher profits.

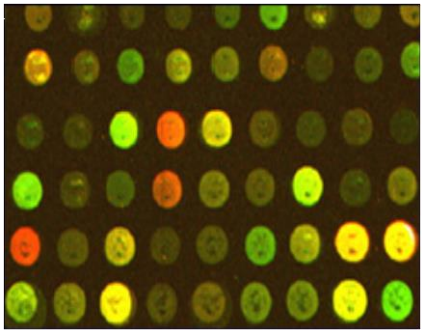
Why Data Mining? Scientific Viewpoint

- Data collected and stored at enormous speeds
  - Remote sensors on a satellite
    - NASA EOSDIS archives over petabytes of earth science data / year
  - Telescopes scanning the skies
    - Sky survey data
  - High-throughput biological data
  - Scientific simulations
    - terabytes of data generated in a few hours

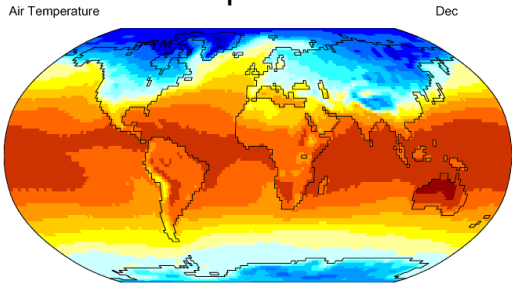


fMRI Data from Brain


Sky Survey Data

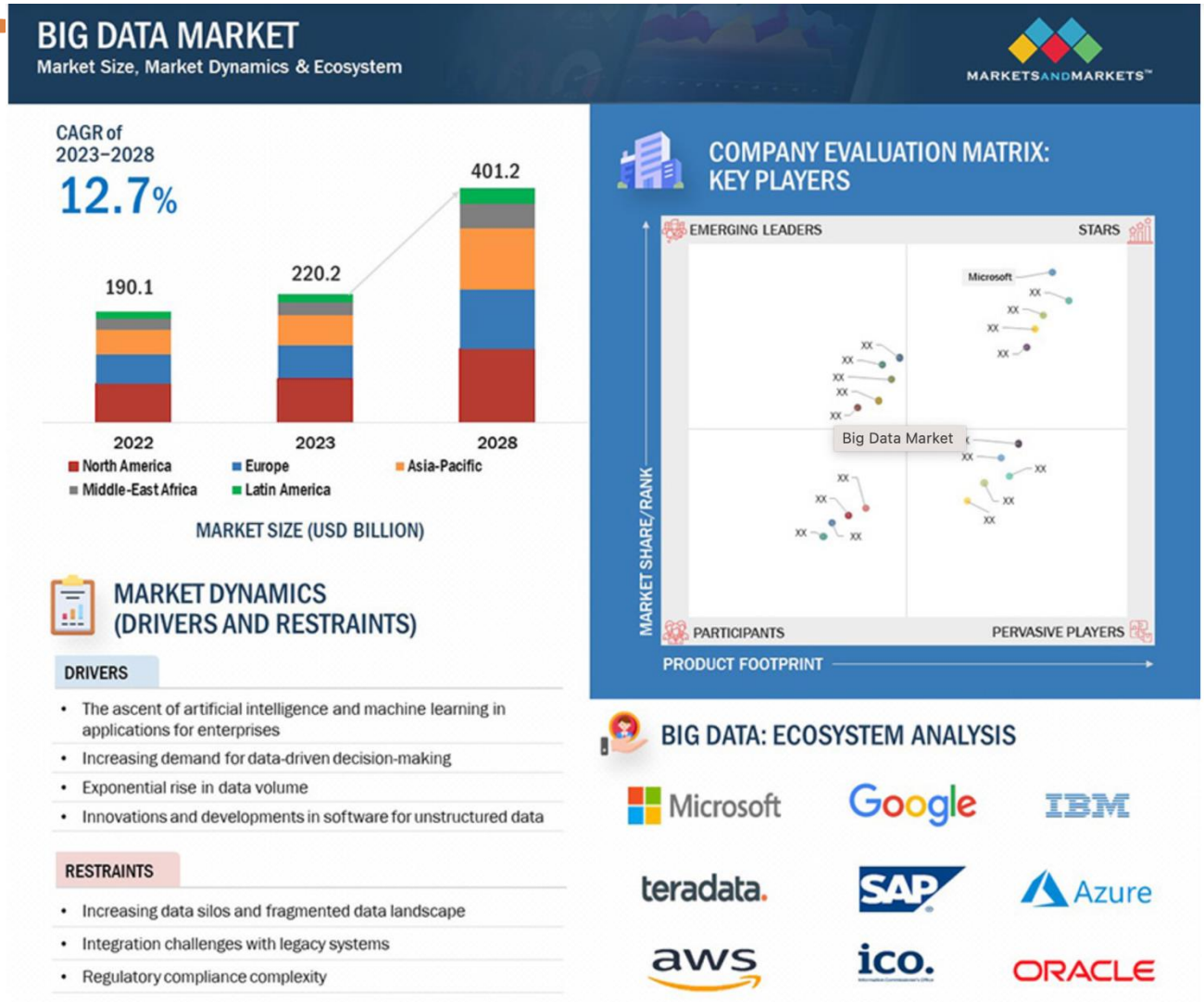

Gene Expression Data

- Data mining helps scientists
  - in identifying patterns and trends
  - in improving efficiency
    - by automated analysis of massive datasets
  - in hypothesis generation.



Surface Temperature of Earth

And a rapidly growing huge market.

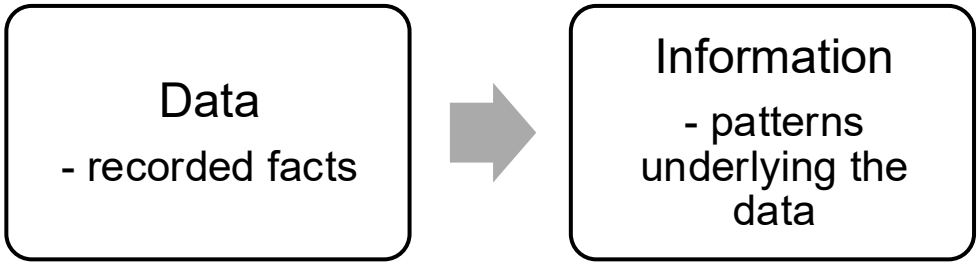


https://www.marketsandmarkets.com/Market-Reports/big-data-market-1068.html

Why Data Mining Matters (Summary)

Why Do We Need Data Mining?

- Massive amounts of data everywhere
  - business, science, healthcare, social media,…
- This data is a potentially valuable resource.
  - But, *Raw data ≠ useful knowledge*
- Humans cannot manually analyze large datasets.
- *Raw data is useless*: need techniques to automatically extract information from it



Data Mining Examples (1)

- Business & E-Commerce
  - Market Basket Analysis
  - Customer Segmentation
  - Targeted advertising



Image credit to: https://tech-talk.org/2015/07/10/data-analytics-for-e-commerce/

Data Mining Examples (2)

- Healthcare & Medicine
  - Disease Risk Prediction
  - Medical Image Mining
  - Personalized Medicine
  - Treatment Effectiveness



Data Mining Examples (3)

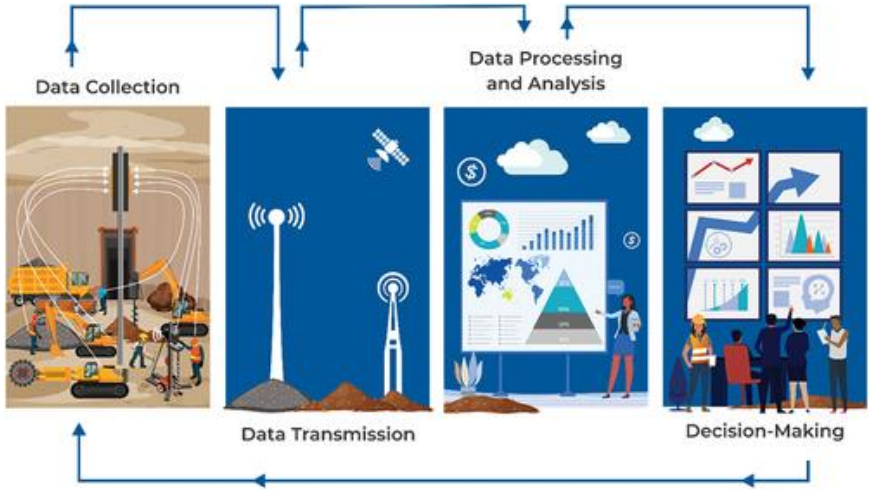
- Web & Social Media
  - Recommendation Systems
  - Social Network Analysis
  - Trend Prediction
  - Personalized Marketing
  - Influencer Identification
  - Sentiment Analysis



Image credit to: https://pragmaetimos.com/social-media-intelligence-and-data-mining-what-is-the-relationship/

Data Mining Examples (4)

- Manufacturing & IoT
  - Predictive Maintenance
  - Quality Control
  - Smart Homes
  - Agriculture
  - etc



https://www.tandfonline.com/doi/full/10.1080/17480930.2024.2347131

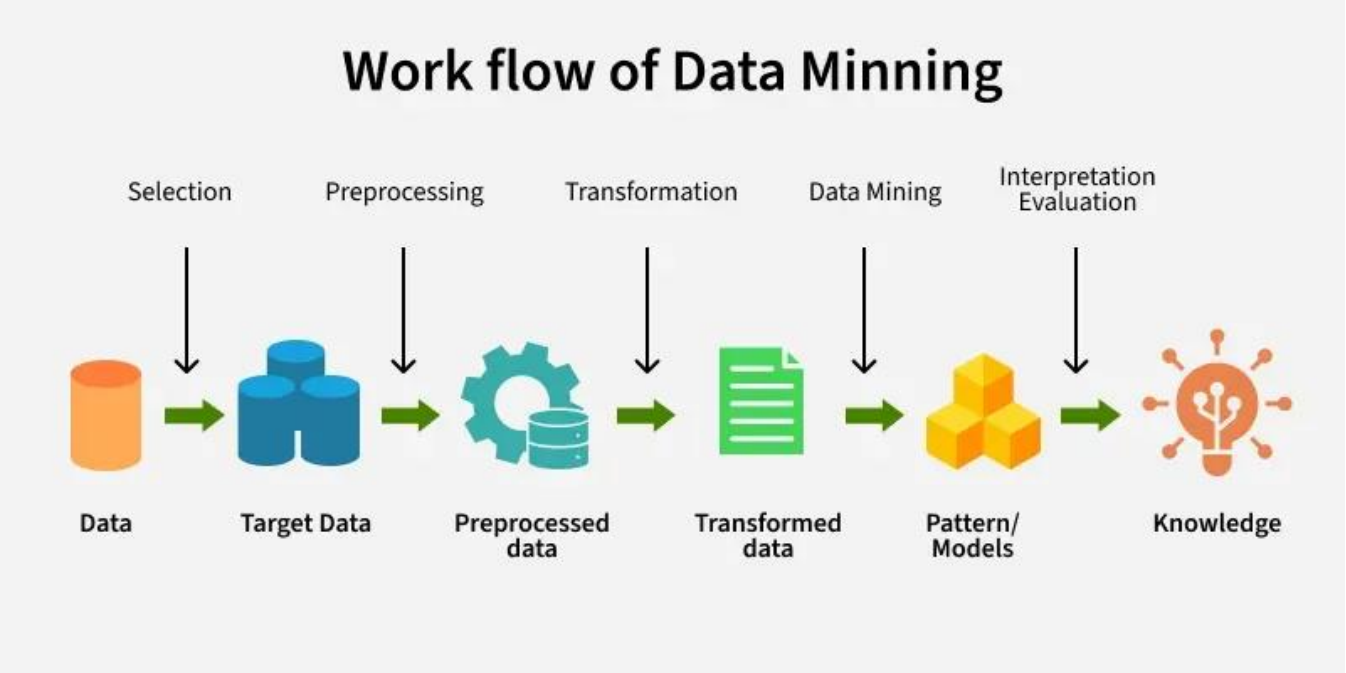
- Science & Environment
  - Climate Pattern Discovery
  - Bioinformatics
  - Drug Discovery
  - Astronomy
  - etc



https://www.linkedin.com/pulse/revolutionizing-environmental-monitoring-data-science-ryan-kmetz-vnt1e

What is Data Mining?

- Many Definitions
  - I. Non-trivial extraction of *implicit*, *previously unknown* and *potentially useful information* from data
  - II. Exploration & analysis, by automatic or semi-automatic means, of large quantities of data in order to *discover meaningful patterns.*

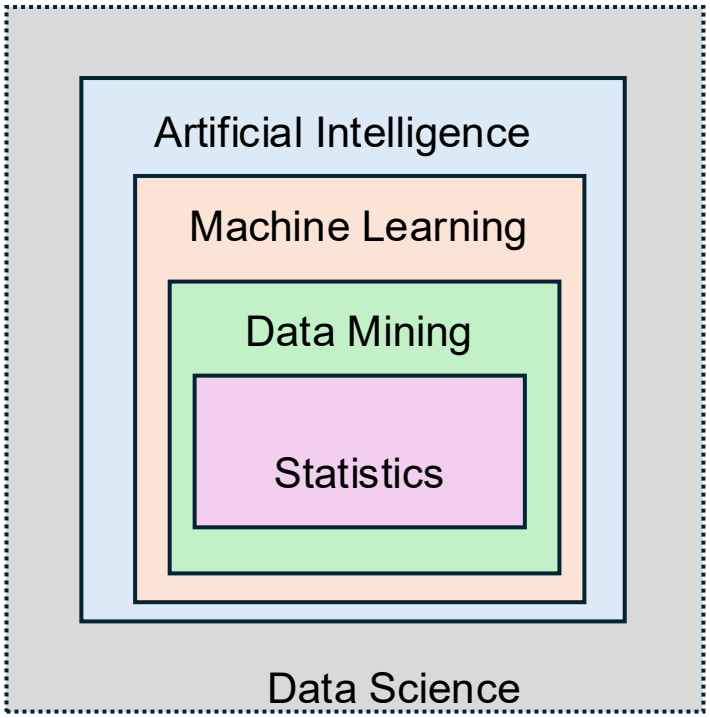


https://www.geeksforgeeks.org/blogs/what-is-data-mining-a-complete-beginners-guide/

Data Mining vs Related Fields (Big Picture)

Data Mining, Data Science, Machine Learning, AI, Statistics

Key idea: These fields are *connected*, not competing.



The goal - build intelligent systems

The engine – algorithms that learn from data

The process – discovering useful patterns from large datasets

The foundation – mathematical tools for inference and uncertainty

The end-to-end practice – turning data into actionable insight.

The Online Retail Example

Problem: Improve sales and customer experience

- Data Science
  - Collect transaction data
  - Clean and preprocess data
  - Visualize and report insights
- Data Mining
  - Discover patterns:
    - "Customers who buy A often buy B"
  - Cluster customers by behavior
  - Detect anomalies or fraud
- Machine Learning
  - Predict customer churn
  - Build recommendation systems
- Statistics
  - Test whether promotions increase sales
  - Estimate confidence intervals
- Artificial Intelligence
  - Intelligent recommendation and pricing systems
  - Automated decision-making

Data Mining is a *core analytical step* inside Data Science that bridges Statistics and Machine Learning to support AI applications.

Types of Data

What Kinds of Data Do We Mine?

- Tabular data (rows & columns)
- Text data
- Time series
- Graphs / networks
- Images & multimedia

This course focus:

- Mainly **structured & semi-structured data**
- With connections to modern methods

Core Data Mining Tasks

What Problems Does Data Mining Solve?

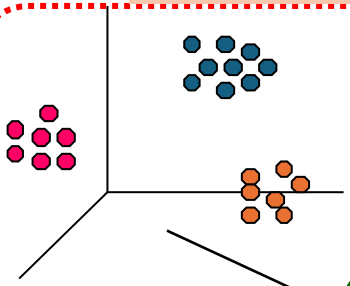
- Classification
- Regression
- Clustering
- Association rule mining
- Anomaly detection
- Dimensionality reduction

These will define **most of them** during our course

Data Mining Tasks …

Descriptive Tasks

Predictive Tasks



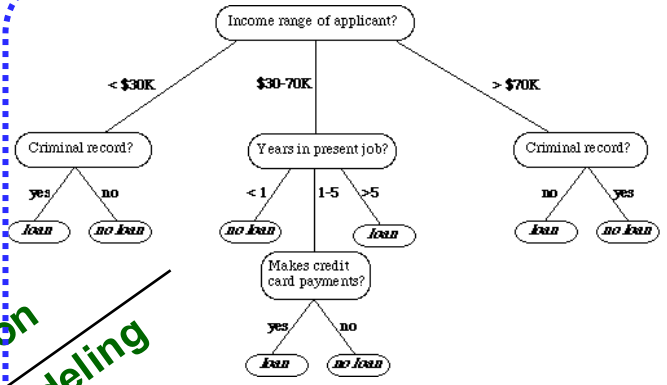
Clustering

Data

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes
11	No	Married	60K	No
12	Yes	Divorced	220K	No
13	No	Single	85K	Yes
14	No	Married	75K	No
15	No	Single	90K	Yes

Regression

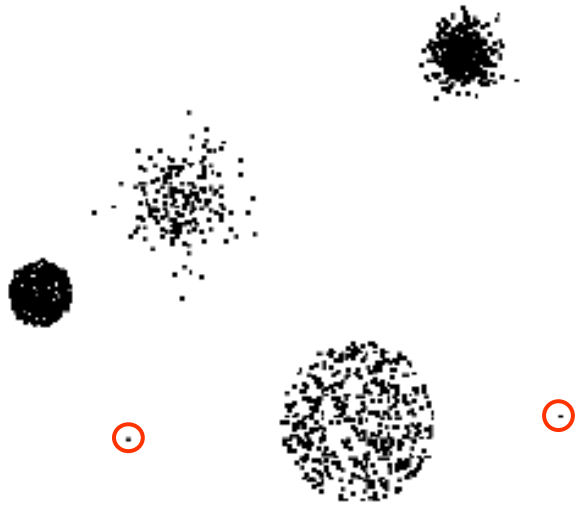
Predictive Modeling



Association Rules

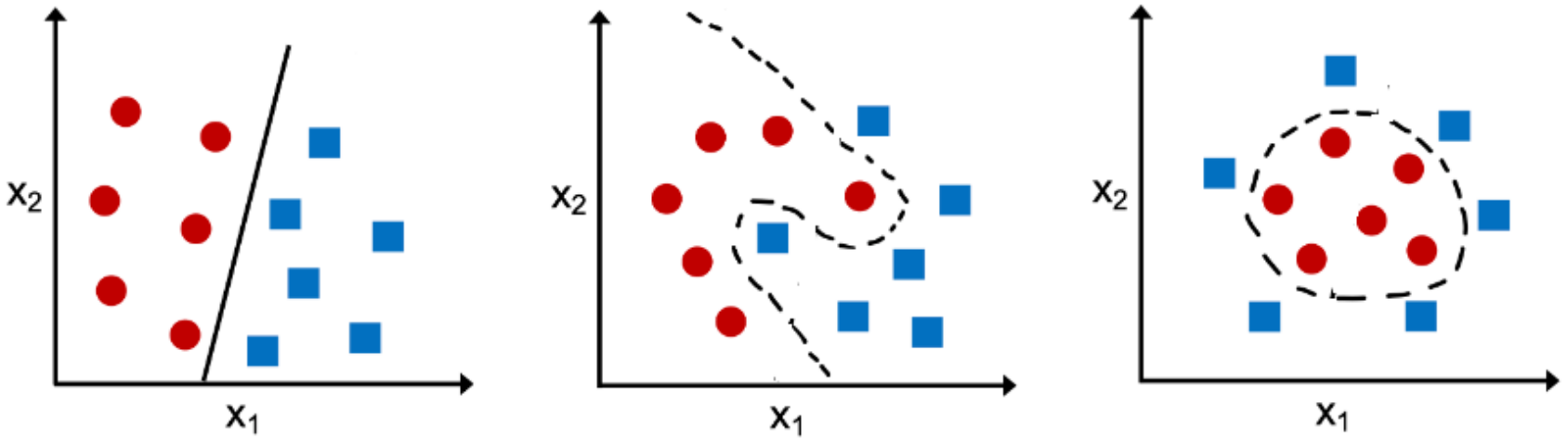


Anomaly Detection



Classification

- Assigning input data to predefined categories or classes.
- It's a supervised learning method where models learn from labeled examples to predict the correct class for new, unseen data, with common types including binary (two classes) and multi-class (multiple classes)



Examples of Classification

- Classifying credit card transactions as legitimate or fraudulent



- Classifying land covers (water bodies, urban areas, forests, etc.) using satellite data
- Categorizing news stories as finance, weather, entertainment, sports, etc
- Identifying intruders in the cyberspace

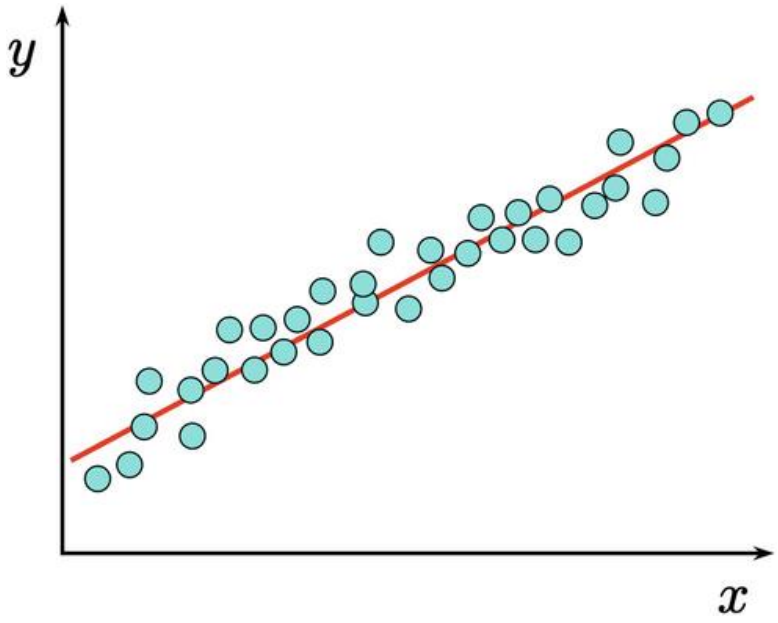


- Predicting tumor cells as benign or malignant
- Classifying secondary structures of protein as alpha-helix, beta-sheet, or random coil

Regression

- Predict a **continuous numerical value** (like price, temperature, or age) based on input features

- It's a supervised learning method where models learn patterns from labeled data to map inputs to real-valued outputs, unlike classification which predicts discrete categories



Examples of Regression

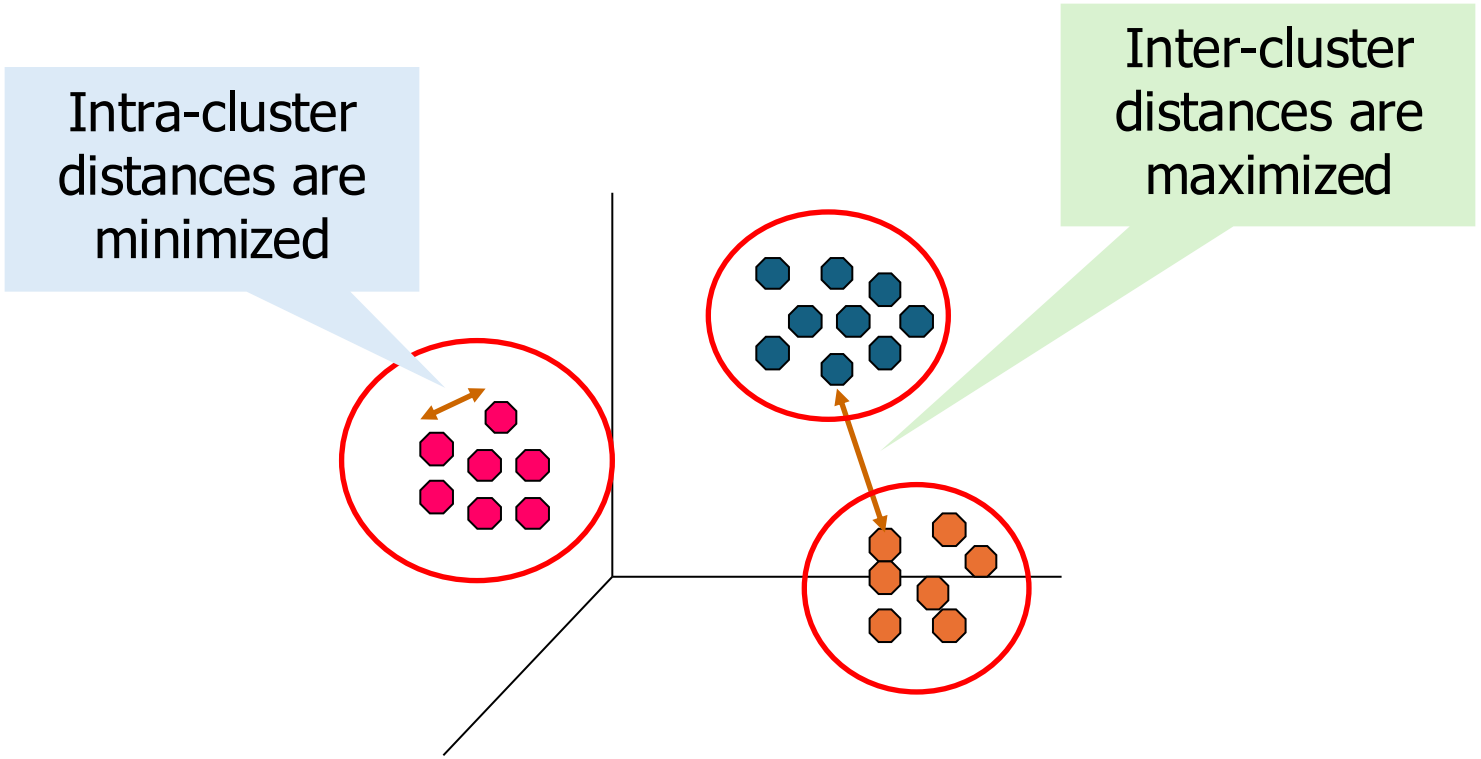
Examples:

- Predicting sales amounts of new product based on advertising expenditure.
- Predicting wind velocities as a function of temperature, humidity, air pressure, etc.
- House Price Prediction
- Time series prediction of stock market indices.
- Traffic Volume Estimation
- Medical Insurance Premium Estimation
- …



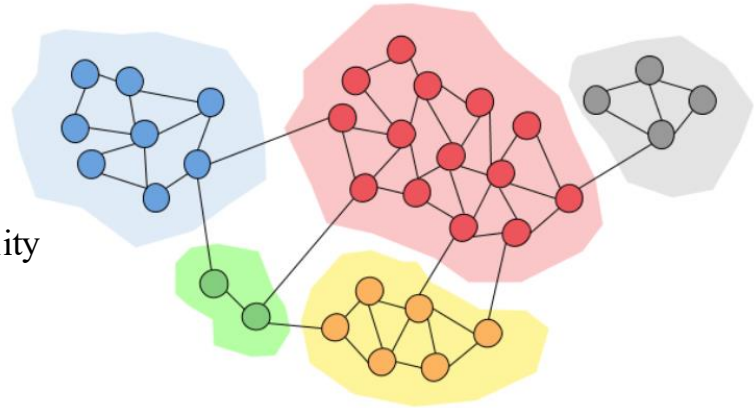
Clustering

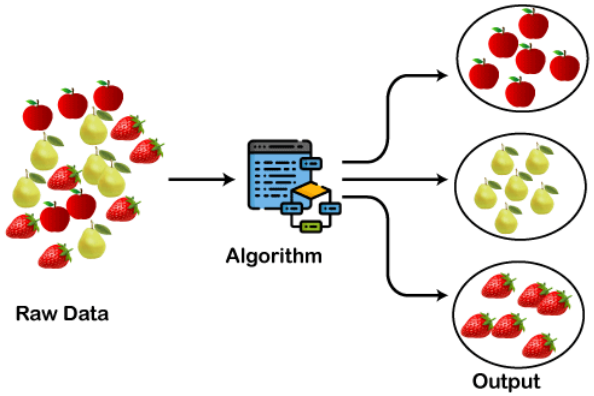
- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups

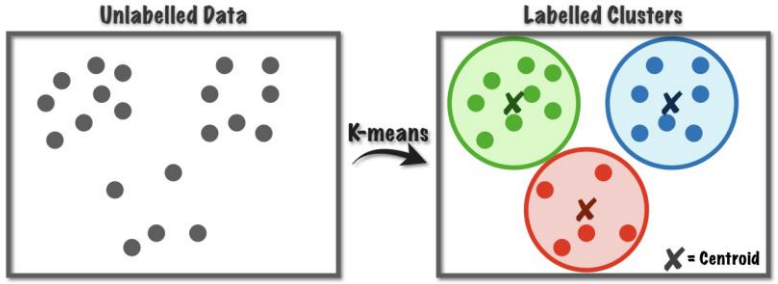


Examples of Clustering

- Some common applications for clustering include the following:
  - Market segmentation
    - Custom profiling for targeted marketing
    - Group stocks with similar price fluctuations
    - social network analysis
  - Medical application:
    - Group genes and proteins that have similar functionality
    - image segmentation
  - Anomaly detection
  - Summarization
    - Reduce the size of large data sets
    - Group related documents for browsing
    - search result grouping

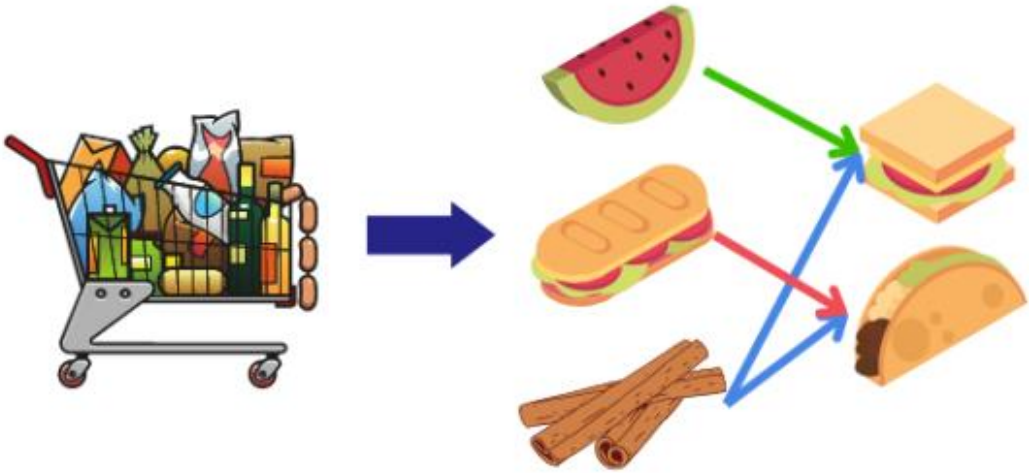






Association Analysis

- Finds interesting relationships between items in large datasets.
- Unsupervised technique that uncovers hidden patterns without needing a predefined target variable.



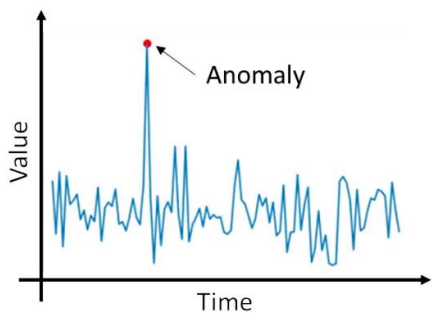
"93% of people who purchased item A also purchased item B"

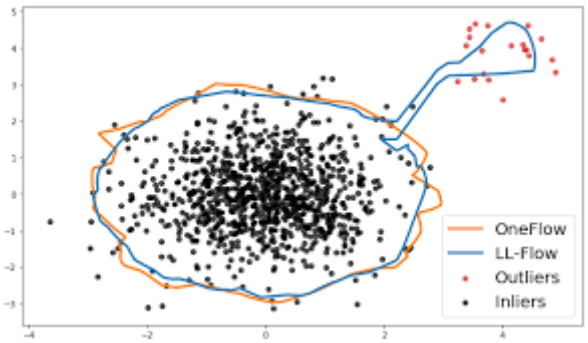
Examples of Association Analysis

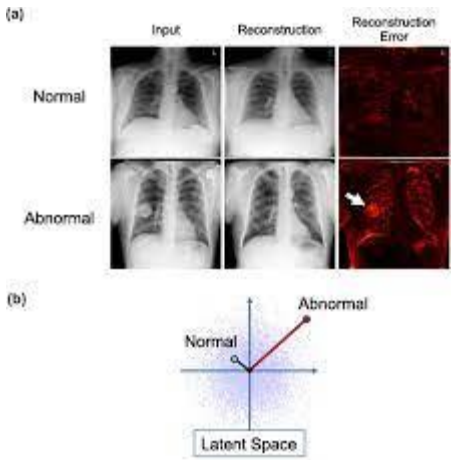
- Market-basket analysis
  - Rules are used for sales promotion, shelf management, and inventory management
- Telecommunication alarm diagnosis
  - Rules are used to find combination of alarms that occur together frequently in the same time period
- Medical Informatics
  - Rules are used to find combination of patient symptoms and test results associated with certain diseases

Deviation/Anomaly Detection

- **Anomaly detection**, sometimes called outlier detection, is a process of finding patterns or instances in a dataset that deviate significantly from the expected or "normal behavior."
- The definition of both "normal" and anomalous data significantly varies depending on the context.
- Applications:
  - Credit Card Fraud Detection
  - Network Intrusion Detection
  - Identify anomalous behavior from sensor networks for monitoring and surveillance.
  - Patient vital sign monitoring
  - Detecting changes in the global forest cover.







Traditional vs Modern Data Mining

Evolution of Data Mining

Traditional:

- Decision trees
- k-means
- SVM
- Logistic regression
- etc

Modern:

- Representation learning
- Deep models
- Graph mining
- Large Language Models (LLMs)
- Vision Language Models (VLMs)
- etc

Why Learn “Traditional” Methods?

Why Not Just Use Deep Learning?

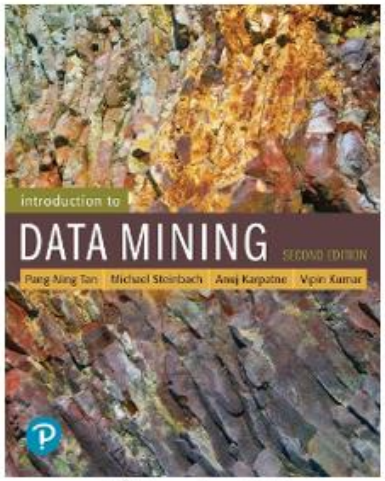
Traditional methods:

- Interpretable
- Data-efficient
- Strong baselines
- Still widely used in industry

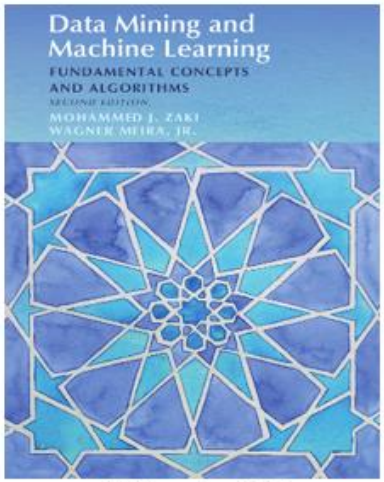
“If you don’t understand the basics, you can’t judge modern models.”

Textbook

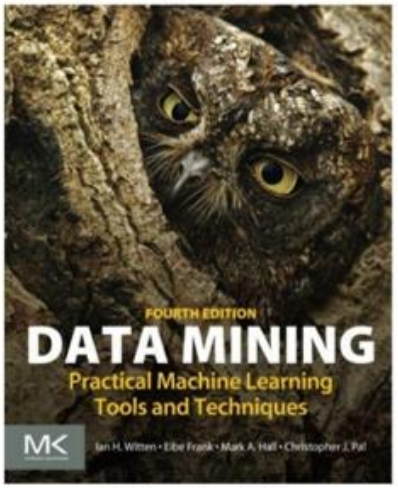
- Introduction to Data Mining (2nd Edition), Tan et al.
- Data Mining and Machine Learning: Fundamental Concepts and Algorithms (2nd Edition), Mohammed et al
- Data Mining: Practical Machine Learning Tools and Techniques (4th Edition) Witten et al.



[Book-A]



[Book-B]



[Book-C]

- Online Materials
- LLMs

Tentative Outline

1. Foundations and Data Understanding

- Introduction to Data Mining
- Data Types & Data Quality
- Data Preprocessing & Feature Engineering
- Similarity, Distance & Evaluation Basics

2. Core Traditional Data Mining Techniques

- Classification (Decision Trees, KNN, Naïve Bayes, Linear Models
- Ensemble Methods
- Clustering (K-means, DBSCAN, GMM)
- Association Analysis
- Dimensionality Reduction

3. Advanced and Modern methods

- From Features to Representations
- Neural Networks for Data Mining
- Graph & Network Mining
- Time Series & Sequential Mining
- Large Language Models & Modern Data Mining
- Vision-Language Models (Multimodal Mining)

4. Applications and Systems Perspective

- Anomaly Detection
- Time Series Analysis
- Domain Applications (i.e., healthcare, social media mining, computer vision + data mining, etc)

5. Projects, Ethics, and Emerging Directions

- Ethics, Fairness
- Privacy & Responsible Data Mining

Grading

Discussion and Participation	10%
Homework Assignments (5)	35%
Exams (2)	30%
Semester Project	30%

Final grade:

89% or more	= A
79% points or more	= B
67% points or more	= C
less than 67%	= F

*Final grade will not be curved..

Course Project

- You will work as a team for semester project, each team has 3~6 members
  (Exceptions of the suggested team members need approval from the instructor)

- Plan:

Deadline	Task
02/19/2026	Form your team, find your group project topic, etc
02/24/2026	3-minute project pitch
05/05/2026	Poster Session
05/10/2026	Group project submission (reports, code, etc)

Late Work and Make-up Tests

- Homework can be submitted after the due date for partial credit, as long as they are submitted before the grades have been posted, for a deduction of 25% if the homework is submitted within 24 hours past the due date.

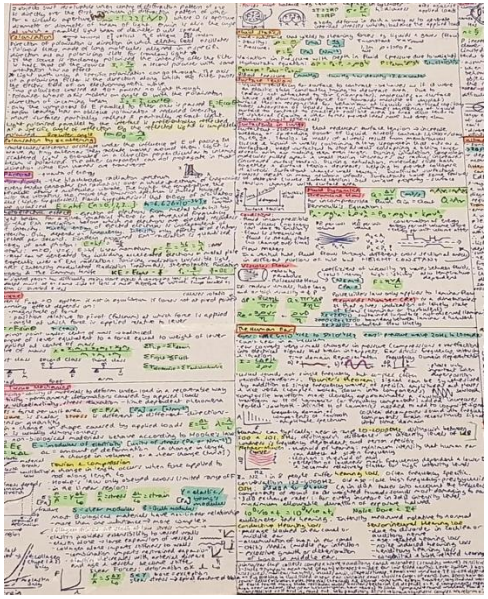
Deadline	Points
Before Deadline	100%
0-24 hours	75%
24-48 hours	50%
48-72 hours	25%
More than 72 hours	0%

- Make-up tests can be scheduled in certain circumstances, such as a health issue with written supporting documentation.

Exams

- Two exams will be scheduled during class time.
- The exams are closed-book, but you are allowed to bring a single A4 sheet of *hand-written notes*, double sided.

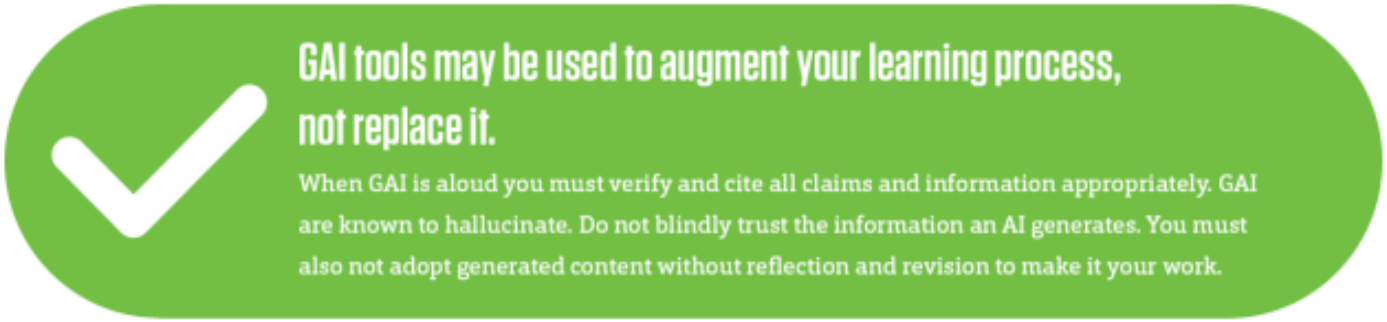
N/A



- Makeup exam may be arranged under specific circumstances, such as a health emergency.
- Accommodations will be provided, but please inform the instructor in advance.

Course Policy: AI tools

- Artificial Intelligence
  - While students are encouraged to use AI tools (such as ChatGPT) to clarify doubts and deepen their understanding of course concepts, utilizing AI to solve or complete homework problems is strictly prohibited. Unauthorized use of artificially generated content violates University Student Academic Standards without consent of the instructor.



- Working together on assignments
  - You are allowed to collaborate on the homework to the extent of formulating your ideas as a group. However, you must write up the solutions to each problem set completely on your own and once your assignment is written up, you must not let others see your solutions. If it has been discovered that you have collaborated with others on your homework or the exams, all of you will *receive a zero and the Academic Dishonesty Policy will apply*.

The use of AI: *Examples*

Exp-1:	Explain to me what SVM is like I'm 7	
Exp-2:	Hi, AI, here is my HW1, could you please provide me with the solution?	
Exp-3:	Hi, AI, could you help me better understand the concepts by providing more examples?	
Exp-4:	Hi, AI, please write a course project report for me. Detailed requirements are listed ….	
Exp-5:	Hi, AI, here is my source code, please revise my code, including debugging and optimization.	

Attendance

- Participation in class discussion and attendance are highly encouraged.
  - Throughout the semester, 12 questions will be posted at random during class sessions. You will need a device (such as a smartphone or laptop) to submit your responses. These questions will count for 10% of your final grade. If you are unable to attend class due to a health issue, please inform me in advance so that I can manually record your participation if a question is posted that day.
- If you do not feel well, you do not need to attend class in person, but please let the instructor know.
- For any respiratory illness, COVID-19 and influenza in particular, the new recommendations suggest returning to normal activities when:
  - Symptoms have been improving overall for at least 24 hour, and
  - If fever is present, no fever for 24 hours without the use of fever-reducing medication.
- Students who are ill are advised to be seen at Student Health Services.

Course Policy

- Student Honor Code and Academic Integrity
  - All students are expected to follow the Honor Code.
  - Student Academic Regulations describes the student standard of conduct relative to the University of Missouri System's Collected Rules and Regulations section 200.010, and offers descriptions of academic dishonesty including *cheating, plagiarism, sabotage, and unauthorized use of artificially generated content*, any of which will be reported to the Vice Provost for Undergraduate Education.
  - Other resources for students regarding academic integrity can be found online.
- Respect for Other
  - You may ask questions or post your opinions of observations of various topics on Canvas. However, you must be respectful of others when you post your comments.
  - Nobody should feel like they are being singled out because of their gender, ethnicity, religion, culture, beliefs, gender representation, sexual orientation, home, family, occupation, military service, age, height, weight, etc. It is a serious matter that can quickly escalate to disciplinary actions. So, please consider how your words could be taken by others.

Questions about Grading

- If you have questions about the grading of assignment,
  - I. please contact the TA first within 7 days after the grade has been released.
  - II. If the issue has not been resolved by the TA, please email TA and Cc to the Instructor, we can resolve the problem together. (with email subject starts with: CS5402-SP26: xxxxx)



If you don't bring your question to the TA or Instructor within the specified time period, no changes will be allowed afterward.

- If you have question about your exams:
  - As required, we must keep your original exam papers for at least three years, so they cannot be returned to you. If you have any questions or would like to review your exam papers, please visit the TA during office hours

Wrap-Up & Questions

Key Takeaways

- Data mining = extracting knowledge from data
- It is interdisciplinary
- Both traditional & modern methods matter
- This course emphasizes **understanding + application**

Questions?

What's Coming Next?

Next Lectures

- Data types & data quality
- Preprocessing & feature engineering
- Distance & similarity
- This is where *real work* starts